

Use-Case Flow Specification

PES University – Dept. of CSE | Lab 1: Requirements Engineering & UML Use-Case Modelling

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Use Case Identification

Use Case: UC-03 — Borrow/Reserve Equipment

Primary Actor: Community Member

Supporting Actors: Library Custodian, Depot Locker System

Preconditions

- The Community Member has a valid, authenticated account in the system.
- The requested equipment item exists in the inventory.
- The requested equipment item is currently available (not checked out).
- The Community Member is not prohibited from borrowing (no overdue items or account restrictions).

Main Success Scenario

1. The Community Member searches for the required equipment by category, name, or keyword.
2. The system displays matching equipment items with current availability status.
3. The Community Member selects an available item to borrow.
4. The system verifies the member's borrowing eligibility (account status, overdue items, borrowing limits).
5. The system checks the applicable loan duration limits for the selected equipment category.
6. The system calculates the required security deposit based on the equipment category.
7. The system displays the deposit amount, loan duration, and expected return date to the member.
8. The Community Member confirms the borrowing request.
9. The system creates a loan record with the member, equipment item, deposit, and return date.
10. The system updates the equipment status to "Checked Out" in the inventory.
11. The Depot Locker System updates the physical availability indicator at the corresponding locker.
12. The system displays a borrowing confirmation to the Community Member.

Alternate Flow

A1 — Member Has Overdue Equipment

At step 4, the system detects that the Community Member has an overdue equipment item.

4a. The system prevents the new borrowing request.

4b. The system displays the details of the overdue item and the associated restriction.

4c. The system informs the member that borrowing cannot continue until the overdue item is returned and any applicable fees are settled.

4d. The use case ends without creating a new loan record.

A2 — Equipment Becomes Unavailable

At step 3, the system detects that the selected item is no longer available.

3a. The system informs the Community Member that the item is no longer available.

- 3b. The system displays current availability and suggests alternatives if available.
- 3c. The Community Member may select another item (return to step 3) or end the use case.

Postconditions

On Main Success Scenario completion:

- A loan record exists associating the member, equipment item, deposit amount, and return date.
- The equipment item status is updated to "Checked Out" in the inventory.
- The member's security deposit is recorded as held.
- The physical availability indicator at the Depot Locker is updated.

On Alternate Flow (A1) completion:

- No new loan record is created.
- The equipment item remains available.
- The member is notified of the restriction and required resolution steps.